

Thinking Different About Wireless

By Kevin Werbach

Open Spectrum Defined

Most wireless frequency bands are licensed, meaning that the government gives an entity such as a radio broadcaster or the military the exclusive right to transmit on those frequencies. That license comes with restrictions on geography, power output, technical characteristics and/or service offerings. Transmission in the band by any other party is prohibited as “harmful interference.” This regime is considered necessary because the alternative would be a “tragedy of the commons”: a chaotic cacophony in which no one could communicate reliably.

The tragedy of the commons idea resonates with our intuitions. After all, too many sheep grazing in the same meadow will use up all the grass. Too many cars on a highway at the same time will cause traffic jams and collisions. Why should spectrum be any different?

Spectrum is different. It is different because it is inherently non-physical, and because technologies developed in recent years make it practical to avoid the tragedy of the commons. What these technologies have in common is that they allow more than one user to occupy the same range of frequencies at the same time, obviating the need for exclusive licensing.

“Open spectrum” is an umbrella term for such approaches. As used here, open spectrum includes established unlicensed wireless technologies such as WiFi. It would be a mistake, however, to conclude that the existence of WiFi proves no further action is needed to facilitate open spectrum. WiFi was designed for short-range data communication and the limitations of current spectrum rules. It therefore still requires wired “backhaul” connections to the public Internet. Moreover, current unlicensed bands and technical standards are not optimized for efficient spectrum sharing. Enlightened policies will allow the emergence of open spectrum systems that are self-contained, and can handle a